

Specific neuromuscular fatigue induced by repetitive isoload concentric knee extension

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1. Introduction

Muscle fatigue is a physiological phenomenon widely discussed in the scientific community. A lot of mechanisms leading to 'a reduction in the ability of the muscle to produce force or power' (Barry and Enoka 2006) have been investigated during the last 20 years. These studies revealed that fatigue processes could affect the central command as well as the muscular effector. However, their contribution depends on task features (Pasquet et al. 2000; Taylor and Gandevia 2008) (e.g. the intensity and duration of exercise or the type of contraction) or on muscle characteristics (particularly fibre-type composition) (Kawakami et al. 2000). If both dynamic and isometric fatiguing contractions have been considered (Babault et al. 2006), concentric exercise effects were mainly tested through isokinetic contractions. Curiously enough, only one study (Klass et al. 2004) used fatiguing weight loading protocol, on the *triceps surae*, to induce neuromuscular fatigue. These authors found that peripheral impairments, particularly excitation–contraction coupling processes, play a major role in the loss of force. Nevertheless, no study had investigated the knee extension movement, an exercise frequently included in rehabilitation protocols of knee injury and strengthening programmes. In this context, our study aimed to characterise the specific neuromuscular fatigue induced by repetitive isoload concentric leg extensions on *quadriceps femoris* (QF).

2. Methods

The experiment was carried out with nine physically active men (age: 23 ± 2.2 years, height: 179.1 ± 6.9 cm, mass: 72.2 ± 8.5 kg). The fatiguing task was performed on a customised isokinetic dynamometer (Biodex[®] System 3 Pro, Shirley, NY, USA), previously validated (Guilhem et al. 2010). It consisted in executing concentric knee extension, kicking against a constant external resistance, equal to 30% of the isometric maximal voluntary

contraction (IMVC), and applied by a plate-loaded resistance training device. During every dynamic contraction, the subjects had to complete a 90° range of motion (ROM), from 100° to 10° (0°, fully extended knee). Moreover, instructions were given to the subjects to extend the leg with the intention of moving the load 'as fast as possible', taking care in respecting the fixed ROM (without isokinetic dynamometer cushioning effect). The return phase (from 10° to 100°) was controlled by the motor of the dynamometer so that the subjects acted only during the knee extension phase. The task was repeated until the subject failed in completing the entire ROM for three of the five last repetitions with a 10° tolerance. In order to qualify neuromuscular induced fatigue and recovery, subjects performed 5-s IMVCs, at a 70° knee flexion before and after the loading set. They were carefully instructed to contract 'as fast and forcefully as possible'. A supra-maximal (+10%) single pulse stimulation (0.5-ms duration, 400-V maximal voltage) was delivered to the femoral nerve during the torque plateau period of each IMVC (interpolated twitch, IT) and, at rest (rest twitch, RT), about 2-s after the end of the contraction. The post-fatigue tests were carried out immediately after the fatiguing task (POST-1), 30-s after POST-1 (POST-2) and 5-m after POST-1 (POST-3). From IMVCs, maximal voluntary rate force development (RFD_{max}) and isometric torque (MVC torque) were assessed, respectively, as the maximal slope and the peak torque of the time–force curve. During these post-fatigue tests, we also determined averaged root mean square (RMS) of the surface electromyographic signal (sEMG) recorded on agonist and antagonist muscles. To highlight a potential central fatigue, activation level (AL) was quantified with the formula:

$$AL(\%) = 100 - (IT \text{ size}/RT \text{ size}) * 100.$$

RT contractile properties, i.e., the maximal rate torque development, twitch peak torque and maximal rate torque relaxation, were computed to specify peripheral fatigue

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degree. From RT sEMG recordings, *M-wave peak-to-peak amplitude* was determined on *vastus lateralis* (VL) muscle and used to normalise VL RMS value (RMS M^{-1} VL ratio) (Duchateau et al. 2002). A one-way ANOVA with repeated measures was used to assess the statistical significance of relative changes between measures over time. Data changes were expressed relative to baseline values.

3. Results and discussion

Our results indicated that the isoload-fatiguing task significantly and durably affects QF performance. MVC torque and RFD_{max} , respectively, decreased by 31.8% and 41.8% ($p < 0.005$) immediately after exercise. In addition, these mechanical parameters were still depressed to a large extent at POST-3 (−19.6%, $p < 0.005$ and −19.5%, $p < 0.01$, respectively). AL and RMS M^{-1} VL ratio values, which are considered as neural drive ratings, supported a lack of central failure ($p = 0.60$ and $p = 0.30$, respectively) after a fatiguing isoload task, as previously reported in the single related study (Klass et al. 2004). However, considering some limits of our investigation, further studies are required to corroborate this result. On the contrary, mechanical performance decline seemed to be mainly explained by muscular disorders. Our results particularly revealed an impairment in muscular action potential propagation, featuring by a significant decrease in *M-wave peak-to-peak amplitude* of the VL muscle (−7.9%, $p < 0.05$). This observation differs from previous results, suggesting that short rest periods between contractions are enough to preserve sarcolemmal excitability (Kawakami et al. 2000; Klass et al. 2004; Babault et al. 2006). Besides, it appears that the intramuscular processes played a major role in decreased performance. Indeed, the loss and recovery of *twitch peak torque* (−46% at POST-1 and still −18% at POST-3, $p < 0.005$) are highly correlated with MVC torque changes ($r = 0.85$). Impaired Ca^{2+} release has been argued recently as the primary mechanism responsible for fatigue-induced reduction in torque but a reduced sensitivity of contractile proteins to Ca^{2+} could not be excluded (Allen et al. 2008). In the same way, the *maximal rate torque development* (electrically induced) was depressed in the same proportion as RFD_{max} (voluntary contraction) ($r = 0.82$). These results vary from those of Klass et al. (2004), in which these authors had observed an accelerated twitch time course of the *triceps suræ*. Indeed, changes in contractile speed might depend on muscle fibre-type composition (Duchateau and Hainaut 1993) and task characteristics. So, in our study, the *maximal rate torque development* decrease could be linked with a higher proportion of fast twitch muscle fibres in QF (i.e. 48% in QF vs. 30% in *triceps suræ*). Lastly, the decline of the

maximal rate torque relaxation values corroborates the assumptions of intracellular Ca^{2+} -controlled processes impairment and muscle-specific results when compared with the outcomes of Klass et al. (2004). Authors agreed to attribute these muscular mechanisms impairments to metabolites accumulation such as Pi, ADP or H^+ (Allen et al. 2008).

4. Conclusion

Although repetitive concentric isoload knee extensions are commonly used in training and rehabilitation protocols, little is known about neuromuscular response of QF to this type of exercise. Our results highlighted significant decreases in QF performance consecutive to an isoload fatiguing task. We did not show any central fatigue, while intracellular Ca^{2+} -controlled processes impairment was suggested as responsible of performance decline. Interestingly, this neuromuscular fatigue profile seems to be characteristic of fatiguing isoload exercise performed on weight-loaded machines, in contrast with outcomes obtained following isometric (Kawakami et al. 2000) or even isokinetic fatiguing task (Babault et al. 2006). These issues call for further studies in order to compare neuromuscular responses to standardised dynamic isoload and isokinetic exhausting exercises.

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